Problem Formulation: RL-Based Learning-Rate Scheduler

Unit 5: Real-World Reinforcement Learning • Project 16 | Target: CIFAR-10 ConvNet-3 | Algorithm: Deep Q-Network (DQN)

1. PROBLEM MOTIVATION & MARKOV DECISION PROCESS (MDP)

Deep neural network optimization is notoriously sensitive to the step size η. Standard heuristic schedulers (e.g., StepLR, Exponential, Cosine Annealing) execute rigid, predefined formulas that cannot adapt dynamically to stochastic gradient shocks, loss plateaus, or saddle points. We formulate dynamic learning-rate adjustment as a closed-loop Markov Decision Process (&S;, &A;, &P;, &R;, γ) where an autonomous RL agent acts as a meta-optimizer over the child CNN training loop.

2. STATE SPACE (&S; ∈ &R;^{6})

At each epoch $t \in \{1, \ldots, T\}$, the state captures the internal optimization dynamics:

$$s_t = [\ \text{TrainLoss}_t,\ \Delta\text{TrainLoss}_t,\ \text{ValAcc}_t,\ \Delta\text{ValAcc}_t,\ (t / T),\ \log_{10}(\eta_t)\ ] \in \&R;^6$$

- **TrainLoss, ΔLoss:** Normalized training loss and gradient momentum velocity.
- **ValAcc, ΔValAcc:** Current generalization performance on held-out validation data.
- **Progress Ratio (t / T):** Temporal horizon position ensuring finite-time awareness.
- **$\log_{10}(\eta_t)$:** Current learning rate in logarithmic scale for scale-invariance.

3. ACTION SPACE (&A;) & BOUNDS

Discrete multiplicative step adjustments clipped to safe operating bounds:

$$\&A; = \{ \times 0.5, \times 0.8, \times 1.0, \times 1.25, \times 2.0 \} \text{ with clipping } \eta_{t+1} \in [10^{-5}, 0.20]$$

4. REWARD SIGNAL (&R;)

Reward directly incentivizes validation gains while heavily penalizing divergence:

$$R(s_t, a_t, s_{t+1}) = 100 \times \Delta\text{ValAcc}_t - 5.0 \times \max(0, \Delta\text{TrainLoss}_t) + R_{\text{terminal}}$$

Where $R_{\text{terminal}} = 10 \times (\text{ValAcc}_T - 0.65)$ if final accuracy exceeds the 65% aspiration threshold.

5. ALGORITHM: DEEP Q-NETWORK (DQN)

- **Q-Network:** MLP [Linear(6, 64) → ReLU → Linear(64, 64) → ReLU → Linear(64, 5)].
- **Experience Replay:** Capacity 2,000 transitions; breaks correlation between consecutive epochs.
- **Target Network:** Hard sync every 2 episodes to stabilize Bellman targets: $y_t = r + \gamma \max_{a'} Q(s', a')$.
- **Exploration:** ε-greedy exploration decaying from 1.0 down to 0.05.

6. EMPIRICAL BENCHMARK RESULTS (CIFAR-10)

Strategy	Policy Type	Final Val Acc	Reward	Convergence Behavior
RL (DQN) Scheduler	Learned Closed-Loop	**52.75%**	**42.70**	Adaptive decay during stagnation
StepLR Heuristic	Rule-based (×0.5 / 5 ep)	55.55%	45.55	Standard practitioner decay
Fixed Learning Rate	Constant (η = 0.01)	56.35%	46.32	Early plateau at local minimum
Random Action Baseline	Uniform Stochastic	49.90%	39.90	High gradient variance, unstable